# MAHIM PATEL

Equity Research Analyst | Quantitative Derivatives Modeler

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NISM-Series-XV Certified (NISM20260000460795-001) · CFA Level 1 Candidate · 7 Years Market Immersion (2019–2026)

## Education

**Nirma University, Institute of Technology** · Ahmedabad, Gujarat — 2024 – 2028

Bachelor of Technology (B.Tech) in Electronics & Instrumentation (EI) — 3rd Year — Cumulative GPA: 8.19 / 10.0

- Coursework: Control Systems Engineering, Digital Signal Processing, Applied Mathematics, Microcontrollers & Embedded Systems, Process Instrumentation.

**Indian Institute of Technology Madras (IIT Madras)** · Chennai, India — 2023 – 2024

Bachelor of Science (BS) in Data Science and Applications — Foundation Certificate (Completed)

- Coursework: Python for Data Science, Statistics & Probability, Discrete Mathematics & Linear Algebra, Computational Thinking & Data Structures.

## Broker-Verified Track Record & Family Portfolio Attribution (FY 2025–26)

**Family Investment Portfolio Management** · Age 13–20 (2019–2026) — Consolidated FY25–26: +INR 3,49,540.51

Cryptographically Verified Zerodha Ledger (console.zerodha.com/verified/32efc750) — Broker-Verified P&L: +INR 2,24,540.51

- Zerodha Broker-Verified Realized P&L (+INR 2,24,540.51): Itemized across Direct IPO Listing Exits (+INR 1,21,759.15: LG Electronics +INR 1,04,494.80 listing-day exit, Coal India +INR 14,264.55, NSDL +INR 2,999.80), Precious Metals ETFs (+INR 85,672.16: Goldcase +INR 75,381.94, Silverbees +INR 10,290.22), and Index ETFs & Equities (+INR 17,109.20: NiftyBees, Coal Eq, Morepen).
- CAMS-Verified Mutual Fund SIP (+INR 77,101.59): +37.87% cumulative gain at 15.23% annualized XIRR (+2.13% p.a. XIRR spread vs. Nifty 50 TRI 13.10%).
- Self-Reported Off-Broker Family Allocations (+INR 1,25,000.00 Net): Captured +INR 1,50,000 on physical silver bullion (2kg family trade) less a disciplined -INR 25,000 stop-loss exit in BTC at $105k (reconciling to +INR 3,49,540.51 consolidated FY25–26 net profit).

## Institutional Equity Research & Valuation Dossiers

**Tamilnad Mercantile Bank Ltd (NSE: TMB)** · 8-Page Initiation Report — October 2026

Banking ALM, Deposit Moat & Excess Return Model (Family Holding: 19.5%) — BUY | Target: INR 1,140 (+30.7%)

- ALM & Asset Quality Forensics (CMP: INR 872.05): Proved TMB's 104-year Nadar merchant deposit base maintains an 86.5% CD ratio (FY26A; 80.2%–81.9% in FY24–25) with zero panic wholesale CD reliance. Evaluated 32.33% CRAR (30.8% Tier-1), 0.69% GNPA, 0.17% NNPA, 96.04% PCR, 2.14% Q1 FY27 RoA, and a -20% Gold LTV shock stress test.
- Blended Valuation: Trades at 1.35x TTM P/B (1.20x FY27E P/B) and 9.5x P/E vs. 2.0x–2.1x P/B at regional peers (CUB/KVB). Blended Justified P/B (50%), Relative P/E (30%), and Residual Income (20%) model yields INR 1,140.00/sh (+30.7% upside).

**BLS International Services Ltd (NSE: BLS)** · 8-Page Initiation Report — October 2026

Segment EBITDA Decomposition & SOTP Model (Family Holding: 22.5%) — BUY | Target: INR 375 (+67.4%)

- Margin & SOTP Analysis (CMP: INR 224.00): Deconstructed optical margin dilution where Digital Services (7.0% EBITDA margin) masks Core Visa EBITDA margins of 40.1% (INR 738 Cr) and INR 3,302 Net Rev/App (+32.9% YoY). Modeled 29.8% ROCE, INR 1,562 Cr net cash (INR 1,617 Cr gross cash less INR 55 Cr debt = INR 37.93/sh), and SOTP valuation with a 17.3% sovereign tender haircut (Bear Case: INR 185.00, -17.4%).

## Quantitative Engineering & Macroeconomic Publications

**Sundial (`NSE_GAMMA_DEX`): Options GEX Terminal** · 10,628 LOC (48 Files) — Python FastAPI + Next.js 14 TS

Dealer Gamma Exposure, Hedge-Band Greeks & Flip Solver (projects.html) — 73.8% Pin Hit Rate ($n = 42$)

- Engineered 5,732-LOC Python FastAPI ASGI backend & 4,896-LOC Next.js 14 terminal ingesting multi-expiry NSE chains via authorized broker APIs & official NSE FO archives. Built 0DTE-safe Discrete Hedge-Band Gamma ($\Gamma_{\text{band}}$ across ±0.5% spot), Put-Call Parity Implied Forwards ($F$) & carry $b = \ln(F/S)/T$, 50-iter Newton-Raphson IV solver, and 2D Vectorized `(401, N)` Zero-Gamma Flip Solver (∼1.5 ms).
- Empirical Validation ($n = 42$ Expiries): Settled within ±0.25% of the primary Hedge-Band GEX Pin Wall in 31/42 Positive-Gamma NIFTY expiries (73.8% hit rate vs. 52.4% static max-OI baseline); recorded 1.84x afternoon range expansion in Negative Gamma.

**Macro Institutional Monographs & Thematic Research** · Research Archive (mahimpatel.com/macro.html) — 2026

- Authored 3 macro monographs (IT Services Agentic AI Token Parity, 300–500 bps Banking Deposit Squeeze, and GIFT City IFSC Corporate BTC Structures) plus a 5-page thematic initiation on Bitcoin Corporate Treasuries (MSTR, SMLR, Metaplanet; ≤15th %ile mNAV rule).

## Core Competencies & Technical Toolkit

- Fundamental Valuation & Financial Modeling: Dynamic 3-Statement Modeling, FCFF/DCF, Sum-of-the-Parts (SOTP), Justified P/B (Gordon Growth), Residual Income (Excess Return), Forensic Accounting Audits, Commercial Banking ALM & Credit-Cost Stress Testing.
- Quantitative Derivatives & Mathematical Finance: Black-Scholes-Merton Greeks ($\Delta, \Gamma, \mathcal{V}, \Theta$), Discrete Hedge-Band GEX ($\Gamma_{\text{band}}$), Put-Call Parity Implied Forwards, Volatility Skew & Surface Interpolation, Statistical Backtesting, Probability, Statistics & Linear Algebra.
- Programming & Systems Engineering: Python (NumPy, SciPy, Pandas, FastAPI ASGI, AsyncIO), TypeScript / Next.js 14, Data Structures & Algorithms, SQL, Git, LaTeX; Digital Asset Microstructure (Saylor Convertible Flywheel, mNAV Accretion, Sec 115BBH, GIFT City IFSC).